

# Business Requirements - Intelligent Invoice Exception Resolution

## Business Objectives & Value Alignment

### The Core Problem

Our Accounts Payable team processes 50,000 invoices monthly, but **30% require manual exception handling** due to discrepancies between invoices and Goods Receipt Notes (GRNs). These exceptions involve:

- **Quantity mismatches:** Invoice shows 105 units, GRN shows 100 units
- **Price variances:** Invoice amount doesn't match PO pricing
- **Missing GRNs:** Invoice arrives before goods are received
- **Partial deliveries:** Multiple GRNs for single invoice

**Current Process (3-4 days per exception):** 1. AP clerk identifies discrepancy in ERP system 2. Manually searches emails for delivery confirmations 3. Contacts supplier via email to clarify discrepancy 4. Waits for supplier response (1-2 days average) 5. Follows up if no response received 6. Coordinates with internal procurement team 7. Makes judgment call on whether to accept variance or reject invoice 8. Documents decision in email threads and spreadsheets

**Why This Is Not Traditional Automation:** - Requires **judgment and context** (5% variance acceptable for Supplier X, not for Supplier Y) - Involves **unstructured communication** (reading/writing emails, understanding intent) - Needs **multi-party coordination** (supplier, procurement, AP manager) - Demands **relationship management** (appropriate tone, timing, follow-up cadence) - Requires **learning from patterns** (Supplier A always ships 2% over, Item B requires manager approval)

### Strategic Objectives

**Primary Goal:** Transform exception handling from manual investigation to **autonomous agent-driven resolution** with human oversight for high-risk decisions.

**Success Metrics:** - Reduce exception resolution time from 3-4 days to **4-8 hours** - Resolve **70% of exceptions autonomously** without human intervention - Improve supplier response rate from 60% to **85%** through better communication - Capture and codify **tribal knowledge** from experienced AP staff - Reduce AP team workload by **40%** to focus on strategic vendor relationships

**Strategic Alignment:** - Supports FY2025 digital transformation roadmap - Enables AP team to shift from transactional work to strategic vendor management - Improves working capital management through faster invoice resolution - Strengthens supplier relationships through faster, more professional communication

## Stakeholder Engagement & Buy-in

**Executive Sponsor:** CFO - Sarah Chen - Strong commitment to AI transformation - Concerned about maintaining vendor relationships during transition - Allocated \$850K budget with expectation of 24-month payback - Quarterly steering committee oversight

### Key Stakeholders:

**AP Manager (Jane Smith)** - 15-person team - **Attitude:** Cautiously optimistic but concerned about job security - **Key Concern:** “Will agents understand the nuances we’ve learned over 10 years?” - **Expectation:** Agents should escalate complex cases, not make risky decisions autonomously

**Procurement Director (Mike Johnson)** - **Attitude:** Enthusiastic supporter - **Key Concern:** Maintaining supplier relationships and contract compliance - **Expectation:** Better visibility into exception patterns to improve procurement decisions

**IT Director (Tom Wilson)** - **Attitude:** Supportive but concerned about complexity - **Key Concern:** Integration with existing ERP, email systems, and maintaining security - **Expectation:** Cloud-native solution with minimal on-premise dependencies

**Compliance Officer (Lisa Brown)** - **Attitude:** Risk-averse, needs assurance - **Key Concern:** Audit trail for autonomous decisions, explainability for financial decisions - **Expectation:** Complete traceability and ability to explain any agent decision

**Supplier Relationship Manager (David Lee)** - **Attitude:** Skeptical about AI handling supplier communication - **Key Concern:** Tone, professionalism, and relationship preservation in agent-generated emails - **Expectation:** Ability to review and approve agent communications before sending (initially)

## Organizational Culture & Innovation

**Innovation Appetite:** Moderate to high - Successfully implemented Workday HCM 2 years ago (6-month project, good adoption) - Deployed RPA for simple data entry tasks (limited success due to brittleness) - Previous AI pilot (chatbot for employee queries) failed due to poor accuracy and lack of context

**Key Cultural Factors:** - **Collaborative decision-making:** Cross-functional teams work well together - **Risk awareness:** Finance culture is conservative, requires strong governance - **Learning mindset:** Monthly continuous improvement meetings, open to experimentation - **Silos exist:** Finance and IT have historically worked independently

**Decision-Making Speed:** 4-6 weeks for major initiatives through governance committees

**Previous Transformation Lessons:** - RPA failed because it couldn't handle exceptions (exactly what we need agents for) - Chatbot failed because it lacked context and couldn't learn from interactions - Workday succeeded because of strong change management and executive sponsorship

## User Adoption & Change Readiness

**Primary Users:** 15 AP clerks - **Age range:** 35-55 years old - **Tech proficiency:** Comfortable with ERP, Excel, email; limited AI exposure - **Current pain points:** Frustrated by repetitive email follow-ups, lost email threads, inconsistent decision-making - **Attitude toward AI:** Mixed - younger staff (35-40) are enthusiastic, senior staff (50+) are skeptical

**Key User Concerns:** 1. **Job security:** "Will agents replace us?" 2. **Loss of control:** "What if agents make wrong decisions?" 3. **Complexity:** "Will this be harder to use than current process?" 4. **Trust:** "How do we know agents are making good decisions?"

**User Involvement Strategy:** - 3 AP clerks (representing different experience levels) on design team - Weekly feedback sessions during development - Pilot with 2 users for 4 weeks before broader rollout - "Agent assistant" framing rather than "replacement"

**Change Fatigue:** Moderate - Workday implementation was 2 years ago, team has recovered

## Change Management Capability

**Change Management Maturity:** Developing (Level 2 of 5) - Have change management methodology (Prosci ADKAR) but inconsistently applied - No dedicated change management team (project managers handle it) - Previous Workday project had strong change management, RPA project had none

### Detailed Change Management Plan:

**Communication Strategy:** - **Pre-Launch** (Months 1-3): - Executive announcement from CFO at all-hands meeting - Weekly email updates on project progress - Dedicated Slack channel (#agentic-ai-transformation) for Q&A - Lunch-and-learn sessions explaining agentic AI concepts - **Pilot Phase** (Months 4-6): - Daily stand-ups with pilot users - Weekly success stories shared company-wide - Monthly town halls with live agent demonstrations - **Rollout** (Months 7-12): - Bi-weekly email updates on adoption metrics - Quarterly steering committee presentations - Recognition program for "agent champions"

**Training Program:** - **Foundational AI Literacy** (1-day workshop, all AP staff): - What is agentic AI and how does it differ from RPA? - How agents make decisions and learn from outcomes - Hands-on demo of agent capabilities - Q&A session addressing concerns - **Agent Supervision Skills** (2-day hands-on training, all AP clerks): - How to review agent decisions and

explanations - When to override agent recommendations - How to provide feedback for agent learning - Escalation procedures and human-in-the-loop workflows - **Advanced Agent Management** (1-week certification, 3 “agent supervisors”): - Prompt engineering and agent configuration - Performance monitoring and drift detection - Troubleshooting common agent issues - Leading continuous improvement initiatives - **Ongoing Learning**: - Monthly lunch-and-learn sessions on new agent capabilities - Quarterly refresher training on best practices - Annual recertification for agent supervisors

**Resistance Management**: - **One-on-one sessions** with skeptical staff (identified: 5 senior AP clerks) - Address specific concerns (job security, loss of control, complexity) - Show career path evolution to “agent supervisor” roles - Emphasize agents handle repetitive work, humans focus on strategic relationships - **Success stories from pilot users**: - Video testimonials from early adopters - Before/after comparisons showing time savings - Recognition and rewards for pilot participants - **Clear career path**: - AP Clerk → Senior AP Clerk → Agent Supervisor → AP Manager - New skills: AI literacy, agent management, data analysis, strategic vendor management - Salary increases tied to new responsibilities (10-15% for agent supervisors) - **Job security guarantees**: - No layoffs due to agent implementation (written commitment from CFO) - 6 FTE reduction through attrition and redeployment to strategic roles - Retraining budget for staff who want to move to other departments

**User Involvement in Design**: - **Design Team** (3 AP clerks representing different experience levels): - Junior clerk (2 years experience): User interface and ease-of-use feedback - Mid-level clerk (5 years experience): Exception handling logic and decision rules - Senior clerk (12 years experience): Tribal knowledge capture and edge case identification - **Weekly feedback sessions** during development (Months 1-6) - **UAT participation** (10 AP clerks, Months 7-8) - **Pilot program** (2 users, Months 9-10)

**Pilot Success Criteria**: - **Go/No-Go Decision** (End of Month 10): - **Minimum thresholds** (all must be met): - 50% autonomous resolution rate - 95% decision accuracy (validated by human review) - 2-day average resolution time (vs 3-4 days current) - 80% user satisfaction score from pilot participants - Zero critical errors (e.g., fraudulent payments, major supplier relationship damage) - **Target thresholds** (2 of 3 must be met): - 70% autonomous resolution rate - 98% decision accuracy - 8-hour average resolution time - **Rollout decision**: - If minimum thresholds met: Proceed with phased rollout - If target thresholds met: Accelerate rollout to full team - If minimum thresholds not met: Extend pilot, address issues, re-evaluate in 4 weeks

**Rollback Plan**: - **Trigger conditions**: - Decision accuracy drops below 90% for 2 consecutive weeks - Escalation rate exceeds 40% (indicates agent struggling) - Critical error occurs (fraudulent payment, major supplier complaint) - User satisfaction drops below 60% - **Rollback procedure**: 1. Immediate pause of agent processing (switch to manual mode) 2. Root cause analysis by IT and AP teams (48-hour deadline) 3. Communicate issue and plan to all stakeholders 4.

Fix and re-test in staging environment 5. Re-approval by AI Ethics Committee before re-deployment 6. Phased re-introduction starting with pilot users - **Manual processing continuity:** - AP team can continue manual exception handling using existing process - All agent data and context preserved for future use - No data loss or process disruption

**Tribal Knowledge Capture Plan:** - **Phase 1: Documentation** (Months 1-2): - Structured interviews with 5 senior AP clerks (2 hours each) - Shadowing sessions to observe decision-making (1 week per clerk) - Review of historical exception resolutions (2 years of data) - Document 200+ supplier-specific rules in structured format - **Phase 2: Validation** (Month 3): - Workshop with full AP team to validate and refine rules - Identify conflicts and edge cases - Prioritize rules by frequency and business impact - **Phase 3: Agent Training** (Months 4-6): - Encode rules into agent knowledge base - Test agent decisions against historical data - Refine rules based on agent performance - **Phase 4: Continuous Learning** (Ongoing): - Agents learn from human corrections and feedback - New patterns automatically added to knowledge base - Quarterly review of learned rules with AP team

**Career Path Evolution:** - **Current AP Clerk Role:** - 80% transactional work (data entry, email follow-ups, manual matching) - 20% judgment work (exception resolution, supplier communication) - **Future “Agent Supervisor” Role:** - 20% agent oversight (review decisions, provide feedback, handle escalations) - 30% strategic vendor management (relationship building, contract negotiations, performance reviews) - 30% process improvement (analyze patterns, identify automation opportunities, refine agent logic) - 20% training and mentoring (onboard new staff, share best practices, lead continuous improvement) - **Skills Development:** - AI literacy and agent management - Data analysis and pattern recognition - Strategic thinking and problem-solving - Vendor relationship management and negotiation - **Compensation:** - AP Clerk: \$80K average - Agent Supervisor: \$90K-\$95K (12-19% increase) - Senior Agent Supervisor: \$100K-\$110K (25-38% increase)

**Supplier Communication Strategy:** - **Notification Approach:** Transparent disclosure - Email to all active suppliers (200+) explaining AI-driven communication - Emphasize benefits: Faster responses, consistent communication, 24/7 availability - Provide opt-out option for suppliers who prefer human-only communication (estimated <5%) - **Communication Content:** - “We’re implementing AI agents to improve our invoice processing efficiency” - “You may receive emails from our AI system for routine exception inquiries” - “All AI communications are reviewed by our AP team” - “For complex issues, you’ll still work directly with our AP staff” - “Contact us if you have any concerns or prefer human-only communication” - **Phased Approach:** - **Phase 1** (Months 1-3): Agent drafts emails, human reviews and sends - **Phase 2** (Months 4-6): Agent sends emails to trusted suppliers (2+ year relationship), human reviews after - **Phase 3** (Months 7+): Agent sends emails autonomously, human reviews only escalations - **Monitoring:** - Track supplier response rates and sentiment - Quarterly supplier

satisfaction surveys - Immediate escalation for any negative supplier feedback

**Success Measurement:** - **User Adoption Metrics:** - % of AP clerks actively using agent system (target: 100% by Month 12) - Average agent interactions per user per day (target: 20+) - User satisfaction score (target: 80/100) - **Change Management Effectiveness:** - Training completion rate (target: 100%) - Resistance incidents (target: <5 over 12 months) - Voluntary turnover rate (target: <10% vs 15% industry average) - **Business Impact:** - Exception resolution time (target: 8 hours vs 3-4 days current) - Autonomous resolution rate (target: 70%) - Decision accuracy (target: 99%) - Supplier satisfaction (target: 85/100 vs 72/100 current)

## Process Maturity & Automation

**Current Process Documentation:** Partially documented - Standard Operating Procedures (SOPs) exist for common scenarios (80% coverage) - **Exception handling is largely tribal knowledge:** - “Supplier X always ships 2% over, just accept it” - “Item Y is high-value, always escalate to manager” - “If it’s end-of-month, be more lenient to hit targets” - “Supplier Z responds faster to phone calls than emails”

**Exception Types Requiring Agent Intelligence** (15,000 exceptions/month):

1. **Quantity/Price Variances** (40% - 6,000/month)
  - Invoice quantity doesn’t match GRN (e.g., 105 units vs 100 units)
  - Price differences from PO (e.g., \$10,500 vs \$10,000)
  - Partial deliveries across multiple GRNs
  - **Agent needs:** Supplier history, contract variance clauses, item criticality assessment, approval thresholds
2. **Missing/Wrong PO Numbers** (25% - 3,750/month)
  - No PO number on invoice (vendor error or PO not created in time)
  - Wrong PO number (vendor uses exhausted or incorrect PO)
  - Multiple POs for single invoice (split orders)
  - **Agent needs:** Search related POs by vendor/date/amount, contact procurement, create emergency PO workflow
3. **Duplicate Invoices** (15% - 2,250/month)
  - Same invoice submitted multiple times (vendor error)
  - Similar charges from different periods (recurring services)
  - Different invoice numbers but same underlying transaction
  - **Agent needs:** Pattern recognition, historical invoice search, fuzzy matching, vendor notification
4. **Timing Issues** (10% - 1,500/month)
  - Early invoicing (goods not yet received, GRN pending)
  - Late submissions (payment terms expired, discount lost)
  - Invoice format problems (non-standard formats, missing fields)
  - **Agent needs:** Delivery tracking, payment term negotiation, format conversion, deadline management

## 5. Data Quality Issues (10% - 1,500/month)

- Missing or incorrect tax details (GST calculation errors)
- Wrong billing address/department (invoice routing issues)
- Incomplete vendor information (bank details, contact info)
- Currency/exchange rate problems (foreign suppliers)
- **Agent needs:** Data validation, vendor master data lookup, tax calculation verification, currency conversion

**Process Complexity:** - **200+ supplier-specific rules** (mostly undocumented tribal knowledge) - **5 approval workflows** based on amount thresholds (\$0-\$10K, \$10K-\$50K, \$50K-\$100K, \$100K-\$500K, >\$500K) - **30% exception rate** requiring judgment calls (15,000 of 50,000 monthly invoices) - **Variable resolution time:** 1 hour for simple cases (duplicate detection), 5+ days for complex disputes (contract interpretation) - **Peak periods:** Month-end (2x volume), quarter-end (3x volume), year-end (4x volume)

**Current Automation Level:** Low (10% automation) - ERP system flags discrepancies automatically (three-way matching) - Email templates exist but require manual customization for each supplier - No automated follow-up or escalation (manual tracking in spreadsheets) - No pattern recognition or learning from historical resolutions - No duplicate detection (relies on AP clerk memory)

**Continuous Improvement Culture:** Reactive - Monthly team meetings to discuss recurring issues (no data-driven analysis) - No systematic pattern analysis or root cause investigation - Process changes happen ad-hoc based on pain points - No feedback loop from resolution outcomes to process improvement - Tribal knowledge not captured or shared systematically

**What Makes This Agentic AI vs Traditional Automation:** - **Context-dependent decisions:** Acceptable variance depends on supplier history, item type, contract terms, timing, business relationship - **Natural language understanding:** Must read supplier emails, understand intent, detect urgency, interpret contract clauses - **Multi-step reasoning:** Investigate → Search history → Retrieve contracts → Communicate → Negotiate → Decide → Escalate → Learn - **Learning from outcomes:** Improve decision-making based on what worked previously, adapt to new supplier patterns - **Relationship management:** Adjust communication style based on supplier relationship (formal for new, casual for trusted) - **Pattern recognition:** Detect duplicates, identify recurring issues, predict exception likelihood - **Autonomous negotiation:** Propose resolutions within boundaries, escalate when needed

## Skills & Capabilities

**Current Team Skills:** - Strong domain expertise in AP processes and supplier management - Proficient in ERP systems (SAP), Excel, email - **No AI/ML experience** in the team - Basic understanding of automation (RPA exposure)

**Skill Gaps:** - Understanding of agentic AI concepts and capabilities - Prompt

engineering and agent interaction patterns - Interpreting agent decisions and explanations - Monitoring agent performance and identifying drift

**Training Plans:** - **Foundational AI literacy:** 1-day workshop for all staff - **Agent supervision skills:** 2-day hands-on training for AP team - **Advanced agent management:** 1-week certification for 3 “agent supervisors” - **Ongoing learning:** Monthly lunch-and-learn sessions on agent improvements

**Partner Strategy:** - AWS Professional Services for initial architecture and implementation - Retain 1 AI consultant for 6 months post-launch for optimization - Knowledge transfer plan to build internal capability

## Success Metrics & Measurement

### Key Performance Indicators:

| Metric                     | Current Baseline | Target    | Measurement Frequency |
|----------------------------|------------------|-----------|-----------------------|
| Exception resolution time  | 3-4 days         | 4-8 hours | Daily                 |
| Autonomous resolution rate | 0%               | 70%       | Weekly                |
| Supplier response rate     | 60%              | 85%       | Monthly               |
| Decision accuracy          | 94%              | 99%       | Weekly                |
| AP team capacity freed     | 0%               | 40%       | Monthly               |
| Vendor satisfaction score  | 72/100           | 85/100    | Quarterly             |

**Baseline Measurement:** - Currently tracked manually through spreadsheet sampling (500 exceptions/quarter) - Need to implement automated tracking as part of solution

**Success Thresholds:** - **Minimum viable:** 50% autonomous resolution, 2-day average resolution time - **Target:** 70% autonomous resolution, 8-hour average resolution time - **Stretch:** 85% autonomous resolution, 4-hour average resolution time

**Measurement Infrastructure Needed:** - Real-time dashboard showing exception status and agent performance - Automated data collection from ERP and agent system - Monthly reporting to steering committee - Quarterly business review with detailed analysis

**Continuous Improvement Approach:** - Weekly agent performance review meetings with AP team - Monthly pattern analysis to identify improvement opportunities (e.g., recurring supplier issues) - Quarterly agent capability enhancements based on learnings (e.g., new exception types, improved decision logic) - Annual strategic review of agentic AI roadmap and expansion to other finance processes - Feedback loop: Human corrections → Agent learning → Improved decisions → Reduced escalations